

IQR Dataset Explanation

IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5IQR	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
LesserRange	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
GreaterRange	322.0	98.35	91.15	88.5	118.75	78.72	390000.0

IQR Analysis – Placement Dataset

The **Interquartile Range (IQR)** is used to measure the spread of the middle 50% of data in the placement dataset. It helps identify normal salary distribution and detect outliers (extreme values).

Formula

$$\text{IQR} = Q3 - Q1$$

Lower Outlier

$$Q1 - 1.5 * \text{IQR}$$

Upper Outlier

$$Q3 + 1.5 * \text{IQR}$$

Explanation of Table

Term	Meaning
IQR	Shows the spread of the middle 50% of student data
1.5 × IQR	Statistical boundary used to identify outliers
Lesser Range	Minimum acceptable value in the dataset
Greater Range	Maximum acceptable value in the dataset

Human Understanding of This Dataset

The salary IQR value is **₹60,000**, which means most students received salaries within this normal variation range.

The **Lesser Range** salary is **₹1.5 Lakhs**. Salaries below this may be considered lower outliers.

The **Greater Range** salary is **₹3.9 Lakhs**. Salaries above these are treated as higher outliers.

Extremely low or high salary packages are considered unusual compared to most students.

Conclusion

The IQR analysis helps identify the normal placement salary range and detect abnormal salary packages in the placement dataset. Most students fall within the standard salary range, while very high or very low salaries are treated as outliers.